The documentation and process conversion measures necessary to comply with this revision shall be completed by 1 December 1994

INCH-POUND

MIL-S-19500/407B
1 September 1994
SUPERSEDING
MIL-S-19500/407A
4 October 1973

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPE 2N3055 JAN AND JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, power, transistor. Two levels of product assurance is provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

Type	P_T		V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_{STG} and T_{OP}	$R_{\Theta JC}$
	T_A = +25°C 1/ 2/	T_C = +25°C 3/							
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N3055	6	117	100	70	7	15	7	-65°C to +200°C	1.5

1/ T_A = Room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750.
2/ Derate linearly 34.2 mW/°C for T_A > +25°C.
3/ Derate linearly 668 mW/°C for T_C > +25°C.

1.4 Primary electrical characteristics.

	h_{FE}	C_{obo}	$\|h_{fe}\|$	$V_{CE(sat)}$	Pulse response	
	I_C = 4 A dc, V_{CE} = 4 V dc 1/	I_E = 0, V_{CB} = 10 V dc, 100 kHz ≤ f ≤ 1 MHz	I_C = 1 A dc, V_{CE} = 4 V dc, f = 100 kHz	I_C = 4 A dc, I_B = 0.4 A dc 1/	t_{on}	t_{off}
		pf		V dc	µs	µs
Min	20		8			
Max	60	700	40	0.75	6	12

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defence Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications, and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-S-19500, and as specified herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

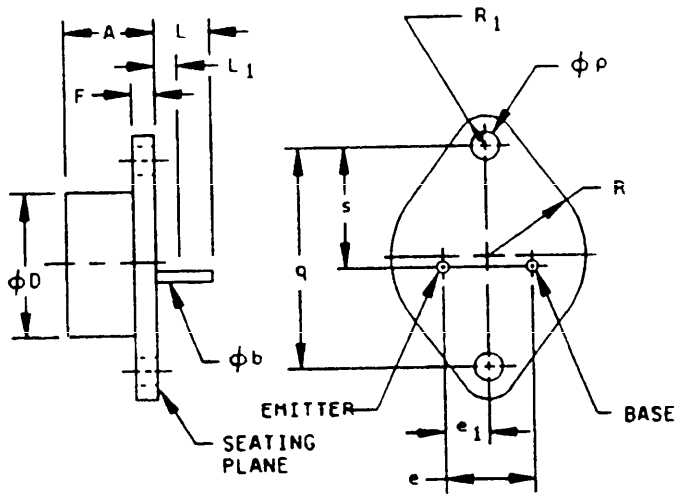
3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.450	6.35	11.43	
ϕb	.038	.043	.97	1.09	5,9
ϕD		.875		22.23	3
e	.420	.440	10.67	11.18	4,10
e_1	.205	.225	5.21	5.72	4,5,10
F	.060	.135	1.52	3.43	
L	.312	.500	7.92	12.70	12,13
L_1		.050		1.27	11
ϕp	.151	.161	3.84	4.09	7
q	1.177	1.197	29.90	30.40	
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	6
s	.655	.675	16.64	17.15	4

FIGURE 1. Physical dimensions.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 inch (6.35 mm) from the seating plane.
4. Measured in the zone .050 inch (1.27 mm) and .250 inch (6.35 mm) from the seating plane.
5. Variations on dimension ϕD in this zone shall not exceed .010 inch (0.25 mm).
6. Outline in this zone is not controlled.
7. When measured in gauging plane .054 +.001, -.000 inch (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 inch (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 shows the preferred method of measurement.
8. The collector shall be electrically connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All 2 leads (see 3.3.1 and 3.3.2).
11. Diameter of leads in this zone is not controlled.
12. For transistor types 2N5415 and 2N5416, dimension L shall be 1.500 inches (38.10 mm) minimum and 1.75 inches (44.4 mm) maximum.
13. For transistor types 2N5415S and 2N5416S, dimension L shall be 0.5 inch (13 mm) minimum and 0.75 inch (19.0 mm) maximum.

FIGURE 1. Physical dimensions - Continued.

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4.3 Screening (JANTX level only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurement shall be made in accordance with table I herein. Devices that exceed the limits of tables I and II herein shall not be acceptable.

Screening (see table II) of MIL-S-19500)	Measurements
	JANTX level
9	Not applicable
11	I_{CEX1} and h_{FE2}
12	See 4.3.1
13	ΔI_{CEX1} = 100 percent of initial value or 100 µA dc whichever is greater; Δh_{FE2} = ±25 percent of initial value; subgroup 2 of table I herein.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = 187.5°C ±12.5°C; V_{CB} = 40 V dc.

NOTE: No heatsink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. End-point electrical measurements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVb of MIL-S-19500.

a. Subgroup 3: Steady-state operation life; method 1027; T_J = 187.5°C ±12.5°C; $V_{CE} \geq 24$ V dc.

b Subgroup 5: Thermal resistance; method 3131; see 4.5.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

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4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2: Terminal strength (tension); method 2036; test condition A; weight = 10 lbs; t = 15s.

b. Subgroup 6: Steady-state operation life; method 1026; T_J = 187.5°C ±12.5°C; $V_{CB} \geq 24$ V dc.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with MIL-STD-750. The maximum limit of $R_{\Theta JC(max)}$ shall be 1.5°C/W. The following test conditions shall apply:

a. I_M: Measurement current . . . . . 10 mA.

b. V_{CE}: Measurements current (same as V_H) . . . . . 30 V.

c. I_H: Collector heating current . . . . . 0.33 A.

d. V_H: Collector-emitter heating voltage . . . . . 30 V.

e. t_H: Heating time . . . . . Steady-state (see MIL-STD-750, method 3131 for definition).

f. t_{MD}: Measurement delay time . . . . . 15 µs.

g. t_{SW}: Sampling window time . . . . . 10 µs maximum.

4.5.3 Coil selection for safe operating area (SOAR) tests. In selecting coils for use in the clamped and unclamped inductive SOAR tests, prime consideration should be given to the recommended commercially available coil. However, due to the extreme critical nature of the coil in these circuits and the wide tolerance of some commercially available coils (+100 percent, -50 percent), it shall be the semiconductor manufacturer's responsibility to prove upon request, compliance or equivalency of any coil used (commercial or in plant designed) to be within (+20 percent, -10 percent) of the specified inductance at the rated current and dc resistance.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage collector to emitter	3011	Bias condition D; I_C = 200 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$	70		V dc
Breakdown voltage collector to emitter	3011	Bias condition B; I_C = 200 mA dc, pulsed (see 4.5.1); R_{BE} = 100 ohms	$V_{(BR)CER}$	80		V dc
Breakdown voltage collector to emitter	3011	Bias condition A; I_C = 200 mA dc, pulsed (see 4.5.1); V_{BE} = -1.5 V dc	$V_{(BR)CEX}$	90		V dc
Collector to emitter cutoff current	3041	Bias condition D V_{CE} = 60 V dc	I_{CEO}		1	mA dc
Collector to emitter cutoff current	3041	Bias condition A V_{BE} = -1.5 V dc V_{CE} = 100 V dc	I_{CEX1}		1	mA dc
Emitter to base cutoff current	3061	Bias condition D V_{EB} = 7 V dc	I_{EBO}		1	mA dc
Forward-current transfer ratio	3076	V_{CE} = 4 V dc I_C = 0.5 A dc	h_{FE1}	40		
Forward-current transfer ratio	3076	V_{CE} = 4 V dc I_C = 4 A dc Pulsed (see 4.5.1)	h_{FE2}	20	60	
Forward-current transfer ratio	3076	V_{CE} = 4 V dc I_C = 10 A dc Pulsed (see 4.5.1)	h_{FE3}	5		
Collector to emitter voltage (saturated)	3071	I_C = 4 A dc I_B = 0.4 A dc Pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.75	V dc
Base emitter voltage (saturated)	3066	Test condition B V_{CE} = 4 V dc I_C = 4 A dc Pulsed (see 4.5.1)	V_{BE}		1.4	V dc
Collector to emitter voltage (saturated)	3071	I_C = 10 A dc I_B = 3.3 A dc Pulsed (see 4.5.1)	$V_{CE(sat)2}$		2	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A V_{CE} = 100 V dc V_{BE} = -1.5 V dc	I_{CEX2}		10	mA dc
Low temperature operation:		T_A = -65° C				
Forward-current transfer ratio	3076	V_{CE} = 4 V dc I_C = 4 A dc Pulsed (see 4.5.1)	h_{FE4}	15		
Subgroup 4						
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 4 V dc I_C = 1 A dc f = 100 kHz	$\|h_{fe}\|$	8	40	
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0 100 kHz ≤ f ≤ 1 MHz	C_{obo}		700	pF
Subgroup 5						
Pulse response	3251	Test condition A				
Turn-on time		V_{CC} = 30 V dc I_C = 4 A dc I_{B1} = 0.4 A dc See figure 2	t_{on}		6	µs
Turn-off time		V_{CC} = 30 V dc I_C = 4 A dc $I_{B1} = -I_{B2}$ = 0.4 A dc See figure 2	t_{off}		12	µs
Subgroup 6						
Safe operating area (continuous dc)	3051	T_C = +25°C; 1 cycle t = 1 s; (see figure 3)				
Test 1		V_{CE} = 7.8 V dc I_C = 15 A dc				
Test 2		V_{CE} = 70 V dc I_C = 1.67 A dc				
Electrical measurements		See table II, steps 1, 2 and 3				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 6 - Continued						
Safe operating area (switching)	3053	Load condition C; (unclamped inductive load) (see figure 4) T_A = +25°C; duty cycle ≤ 10 percent, R_S ≤ 0.1 ohms				
Test 1		t_p = 5 ms (vary to obtain I_C); R_{BB1} = 2 ohms; V_{BB1} ≥ 10 V dc; R_{BB2} = 100 ohms; V_{CC} ≥ 10 V dc; V_{BB2} = 1.5 V dc; I_C = 15 A dc; the coil used shall provide a minimum inductance of 2 mH at 15 A dc with a maximum dc resistance of 0.1 ohm (for reference only: Signal Transformer Co. CH-50, or equivalent) (see 4.5.3)				
Test 2		t_p = 20ms (vary to obtain I_C); R_{BB1} =30 ohms; V_{BB1} ≥ 10 V dc; R_{BB2} = 100 ohms; V_{CC} ≥ 10 V dc; V_{BB2} = 1.5 V dc; I_C = 3.8 A dc; the coil used shall provide a minimum inductance of 40 mH at 3.8 A dc with a maximum dc resistance of 0.5 ohm (for reference only: Signal Transformer Co. CH-8, or equivalent) (see 4.5.3)				
Electrical measurements		See table II, steps 1, 2, and 3				
Subgroup 7						
Safe operating area (switching)	3053	T_A = +25°C; duty cycle ≤ 10 percent, t_p = 10 ms (vary to obtain I_C); R_S ≤ 0.1 ohms; V_{CC} ≥ 10 V dc; I_C = 15 A dc; clamp voltages = 90 V dc (see figure 5)				
Electrical measurements		See table II, steps 1, 2, and 3				

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups A, B, and C electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition A V_{BE} = -1.5 V dc V_{CE} = 100 V dc	I_{CEX1}		1	mA dc
2.	Emitter to base cutoff current	3061	Bias condition D V_{EB} = 7 V dc	I_{EBO}		1	mA dc
3.	Forward-current transfer ratio	3076	V_{CE} = 4 V dc I_C = 4 A dc Pulsed (see 4.5.1)	h_{FE2}	20	60	
4.	Collector to emitter cutoff current	3041	Bias condition A V_{BE} = -1.5 V dc V_{CE} = 100 V dc	ΔI_{CEX1}	±100 percent of initial value or 100 µA dc, which ever is greater		
5.	Forward-current transfer ratio	3076	V_{CE} = 4 V dc I_C = 4 A dc Pulsed (see 4.5.1)	Δh_{FE2}	±25% change in initial recorded value		

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, and 3.

b. Subgroup 3, see table III herein, steps 1, 2, 3, 4, and 5.

c. Subgroup 6, see table III herein, steps 1, 2, 3, 4, and 5.

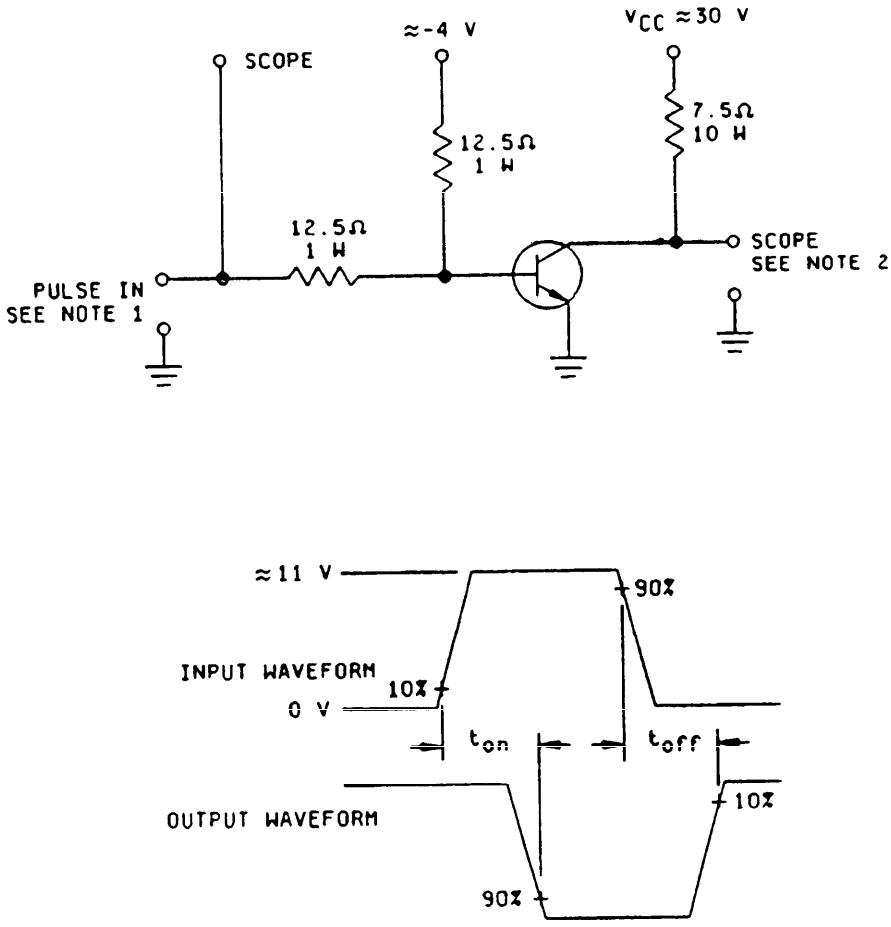
2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, and 3.

b. Subgroup 3, see table III herein, steps 1, 2, and 3.

c. Subgroup 6, see table III herein, steps 1, 2, 3, 4, and 5.

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NOTES:
1. The rise time (t_r) and fall time (t_f) of applied pulse shall be each ≤ 20 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50Ω; pulse width = 20 μs.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 50 pF; rise time ≤ 2 ns.

FIGURE 2. Pulse response test circuit.

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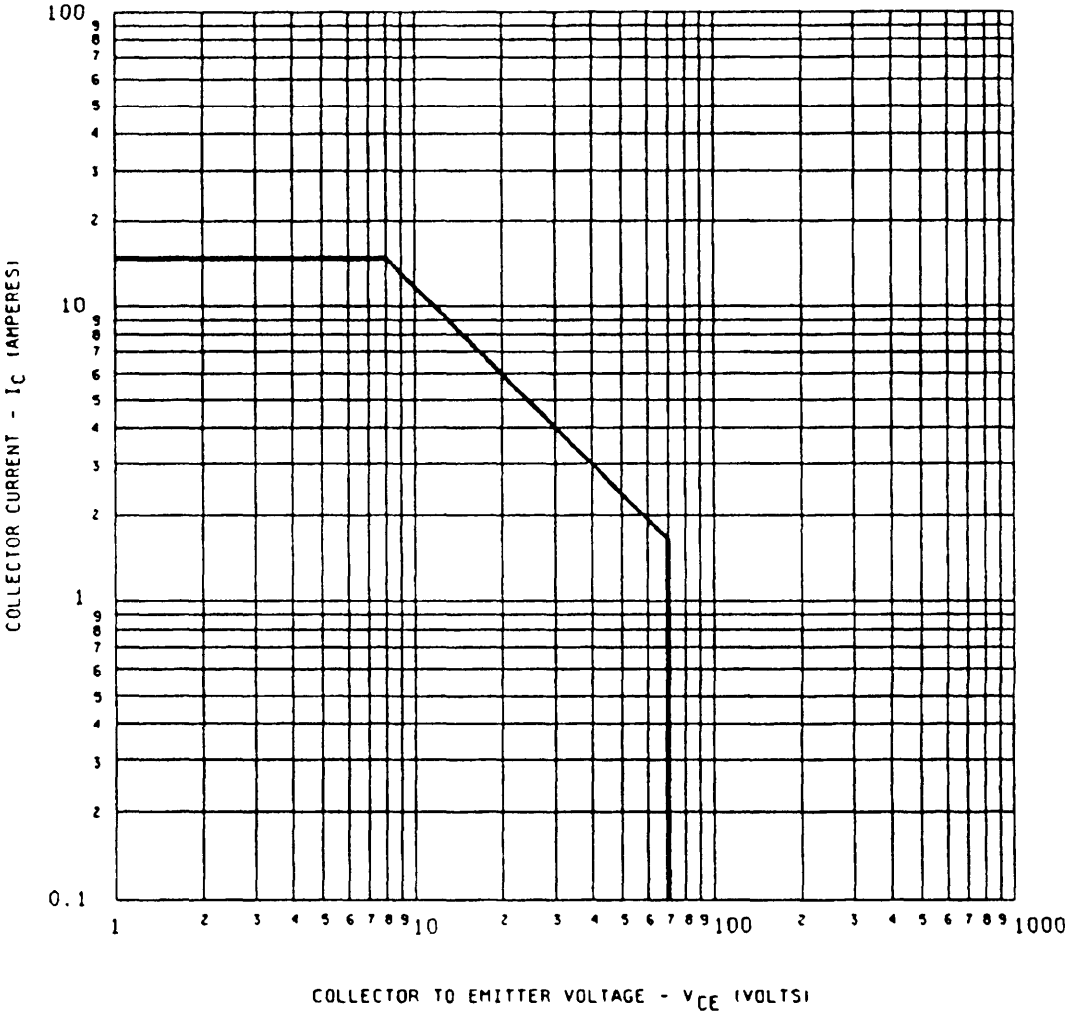


FIGURE 3. Maximum safe operating area graph (continuous dc).

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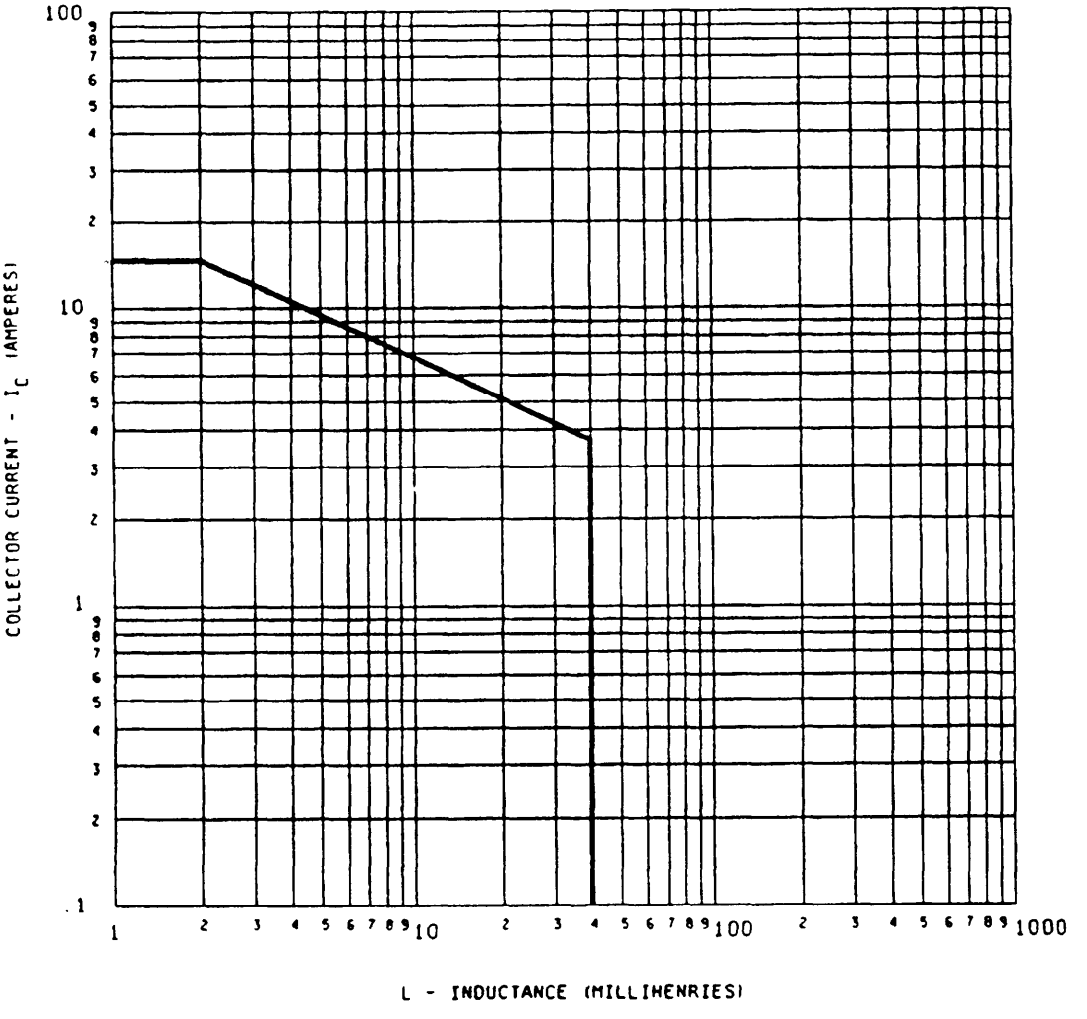
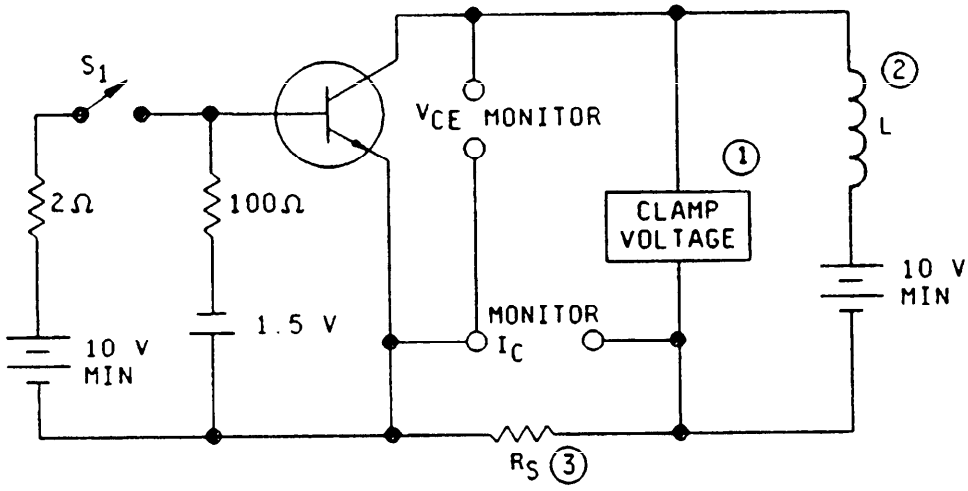


FIGURE 4. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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NOTES:
1. Either a clamping circuit or clamping diode may be used.
2. The coil used shall provide a minimum inductance of 5 mH at 15 A dc with a maximum dc resistance of 0.1 ohm. (For reference only: Signal Transformer Co., CH-20, or equivalent). (See 4.5.3).
3. $R_S \leq 0.1$ ohm, 12 W, 1 percent tolerance maximum (non-inductive).

Procedure:
1. With switch S1 closed, set the specified test conditions.
2. Open S1. Device fails if clamp voltage not reached and maintained until the current returns to zero.
3. Perform specified end point tests.

FIGURE 5. Clamped inductive sweep test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type designation and quality assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AV, MI, MU, SM
Navy - AS, CG, MC
Air Force - 13, 15, 19, 80, 85, 99
DLA - ES

Preparing activity:
DLA - ES

Agent:
DLA - ES

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/407B	2. DOCUMENT DATE (YYMMDD) 940901

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER TYPE 2N3055 JAN AND JANTX

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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